To illustrate the use of off-chain and blockchain-based on-chain data, let’s revisit the airline system consortium (ASK) use case from chapter 2 and complete it with a simple web UI, an off-chain data store, and an updated smart contract. The smart contract design in this implementation also illustrates the design guideline for on-chain data definition.

6.4 ASK airline system

ASK is a decentralized airline consortium creating a marketplace for unused seats on airlines. You were introduced to ASK in chapter 2, with a detailed design description, but the solution was not fully developed because it required you to know more blockchain concepts. Now you are ready to tackle the ASK development task. ASK is not a new idea, but this novel implementation offers an opportunity for newer business models involving the recording capabilities of the blockchain.

ASK illustrates many blockchain concepts, including

- Coexistence of a traditional centralized system and a decentralized blockchain system as parts of a larger system (a concept discussed in chapter 1)
- Off-chain and on-chain data (the topic of this chapter)
- Use of cryptocurrency (tokens) while keeping your fiat currency (say, dollars) for regular operations (covered in chapter 9)
- End-to-end Dapp development using the Truffle IDE (introduced in chapters 4 and 5)

6.4.1 ASK concept